



BAULKHAM HILLS HIGH SCHOOL

2016
YEAR 11 YEARLY

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- All necessary working should be shown in every question
- Marks may be deducted for careless or badly arranged work
- Attempt all questions

Total marks – 70

This paper consists of TWO sections.

Section 1 – Multiple Choice **10 marks**

Section 2 – Extended Response **60 marks**

Attempt all questions

Start a new page for each question

Section 1 –Multiple Choice (10 marks)

Attempt all questions.

Answer the following on the booklet provided.

- 1 A yacht sailed from A to B on a bearing of $196^\circ T$. To sail from B directly back to A, the bearing it should travel is:

(A) $016^\circ T$ (B) $074^\circ T$ (C) $164^\circ T$ (D) $196^\circ T$

- 2 Simplify $\frac{1}{5-\sqrt{3}}$

(A) $\frac{5+\sqrt{3}}{2}$ (B) $\frac{5+\sqrt{3}}{22}$ (C) $\frac{5-\sqrt{3}}{2}$ (D) $\frac{5-\sqrt{3}}{22}$

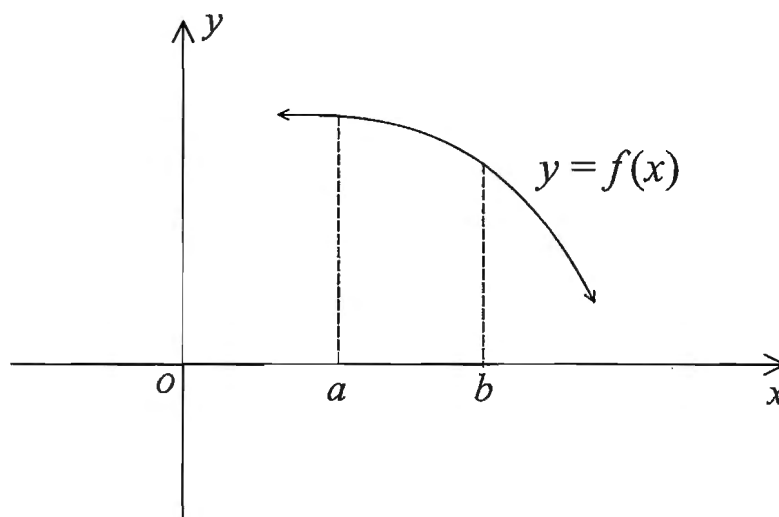
- 3 If α and β are the roots of $5x^2 + 15x - 3 = 0$, then $\alpha^2 + 3\alpha\beta + \beta^2$ is :

(A) $\frac{42}{5}$ (B) $-\frac{42}{5}$ (C) $\frac{48}{5}$ (D) $-\frac{48}{5}$

- 4 Correct to the nearest degree, the angle of inclination from the positive x -axis of a line with x -intercept 2 and y -intercept $2\sqrt{2}$ is:

(A) 35° (B) 55° (C) 125° (D) 155°

- 5 For the function $y = f(x)$ and the domain $a \leq x \leq b$, which of the following is true?



(A) $f'(x) > 0$ and $f''(x) < 0$ (B) $f'(x) < 0$ and $f''(x) < 0$
(C) $f'(x) > 0$ and $f''(x) > 0$ (D) $f'(x) < 0$ and $f''(x) > 0$

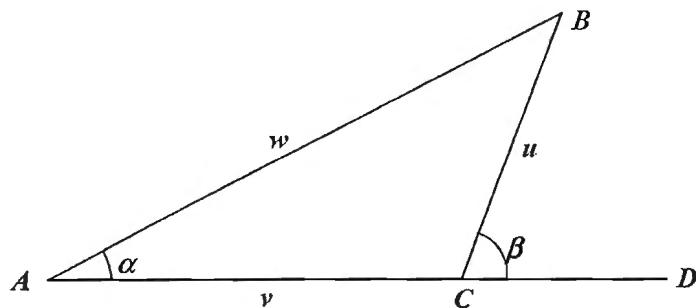
- 6 Which of the following expressions is the correct simplification of $\frac{\operatorname{cosec} \theta \sec \theta}{\tan \theta}$?

(A) $\sin^2 \theta$ (B) $\cos^2 \theta$ (C) $\sec^2 \theta$ (D) $\operatorname{cosec}^2 \theta$

7 The solutions of $\sqrt{3}\tan x + 1 = 0$ for the domain $0^\circ \leq x \leq 360^\circ$ are:

- (A) $x = 30^\circ, 210^\circ$ (B) $x = 150^\circ, 210^\circ$
 (C) $x = 150^\circ, 330^\circ$ (D) $x = 210^\circ, 330^\circ$

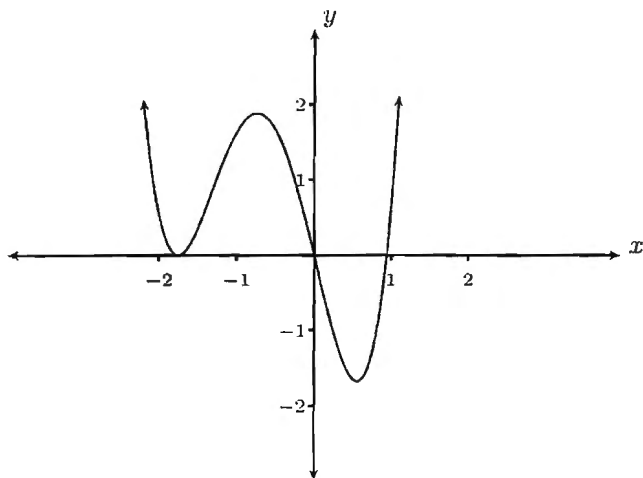
8



In the diagram above, ACD is a straight angle. Which of the following statements is true?

- (A) $w^2 = u^2 + v^2 - 2uv\cos\beta$
 (B) $w^2 = u^2 + v^2 + 2uv\cos\beta$
 (C) $u^2 = v^2 + w^2 - 2vw\cos\beta$
 (D) $\frac{u}{\sin\alpha} = \frac{w}{\cos\beta}$

9



The diagram above shows the function $y = f(x)$.

A student wishes to solve the equation $f(x) = 2 - |x|$ graphically. The correct number of solutions is:

- (A) 1 (B) 2 (C) 3 (D) 4

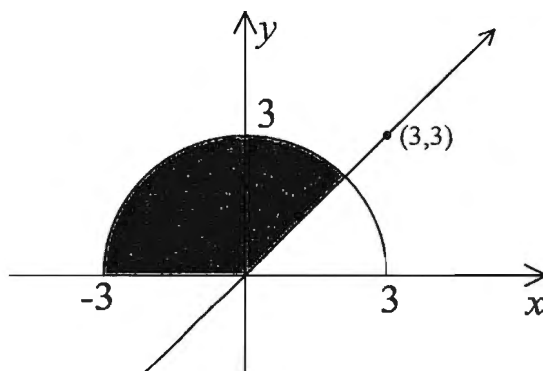
10 Given $f(x + 1) = x^2 - x + 1$, then $f(x)$ in its simplest form is:

- (A) $x^2 + x + 1$ (B) $x^2 + x + 3$
 (C) $x^2 - 3x + 3$ (D) $x^2 - 3x + 1$

End of Section I

Section II – Extended Response**All necessary working should be shown in every question.**

	Marks
Question 11 (15 marks) - Start on the appropriate page in your answer booklet	
a) Simplify $\frac{x+3}{5} - \frac{x-2}{4}$	2
b) Solve $ 3x - 2 = 7$	2
c) Factorise $6x^2 + 11x - 10$	2
d) For the function $f(x) = \sqrt{x^2 - 16}$:	
(i) Find the domain and range	2
(ii) Is the function odd, even or neither? (Justify your answer by showing working)	2
e) Write a set of inequalities which define the shaded region.	2



- f) Find the solutions of the equation $2 \cos^2 \theta = 3 \sin \theta + 3$, for $0^\circ \leq \theta \leq 360^\circ$. 3

End of Question 11

Question 12 (15 marks) - Start on the appropriate page in your answer booklet**Marks**

- a) Simplify $\frac{3^{2n}-1}{6^n-2^n}$ 3
- b) Differentiate with respect to x
- (i) $\frac{4}{1-x}$ 2
- (ii) $2x\sqrt{3x+1}$ 2
- (iii) $\frac{4x-1}{2x+1}$ 2
- c) Evaluate, showing all working:
- $$\lim_{x \rightarrow -2} \frac{x^3 + 8}{2x + 4}$$
- 2
- d) Find the equation of the tangent to $y = x^2 - 4x$ at the point $(1, -3)$ on it. 2
- e) Solve $|2x - 3| = 4x + 1$ 2

End of Question 12

Question 13 (15 marks) - Start on the appropriate page in your answer booklet

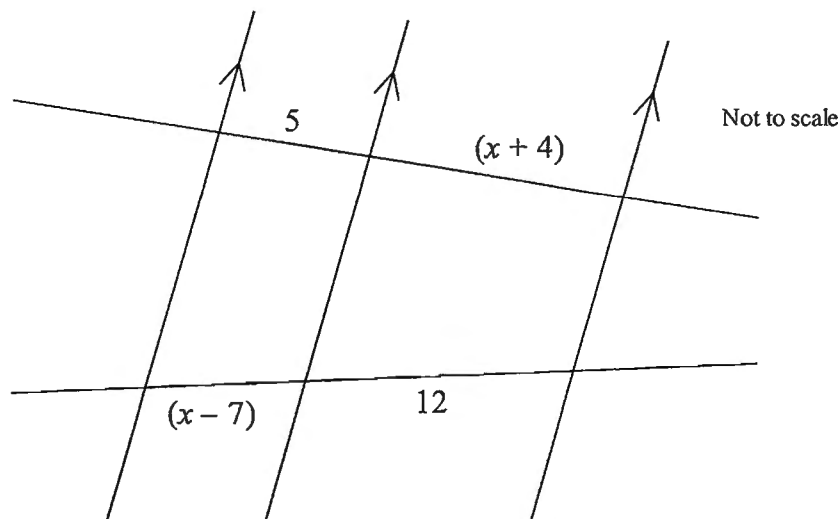
Marks

- a) Find the exact value of $\cos \theta$, given that $\tan \theta = 7$ and $\sin \theta < 0$.

2

- b) Find the value of x .

2



- c) If $6x^2 - 11 \equiv A(x + 2)^2 + Bx + C$, find the values of A , B and C .

3

- d) For which values of k does the equation $x^2 - 2x + 3 - k = 0$ have real roots?

2

- e) Consider the function $f(x) = \frac{1}{2}x^4 - 2x^3 + 2$.

(i) Find the coordinates of the stationary points of $y = f(x)$.

2

(ii) Determine the nature of these stationary points.

2

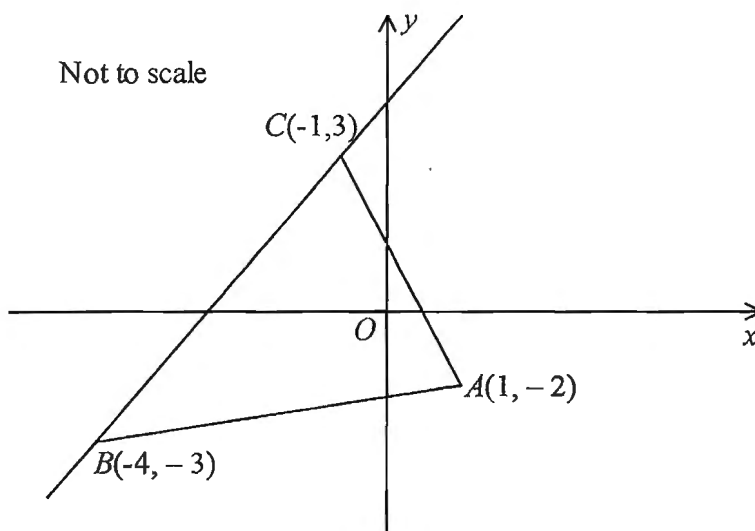
(iii) Sketch the graph of $y = f(x)$, clearly showing the stationary points and y -intercept. (You need not find the x -intercepts).

2

End of Question 13

Question 14 (15 marks) - Start on the appropriate page in your answer booklet

- a) The diagram shows the points $A(1,-2)$, $B(-4,-3)$ and $C(-1,3)$, and the line BC .



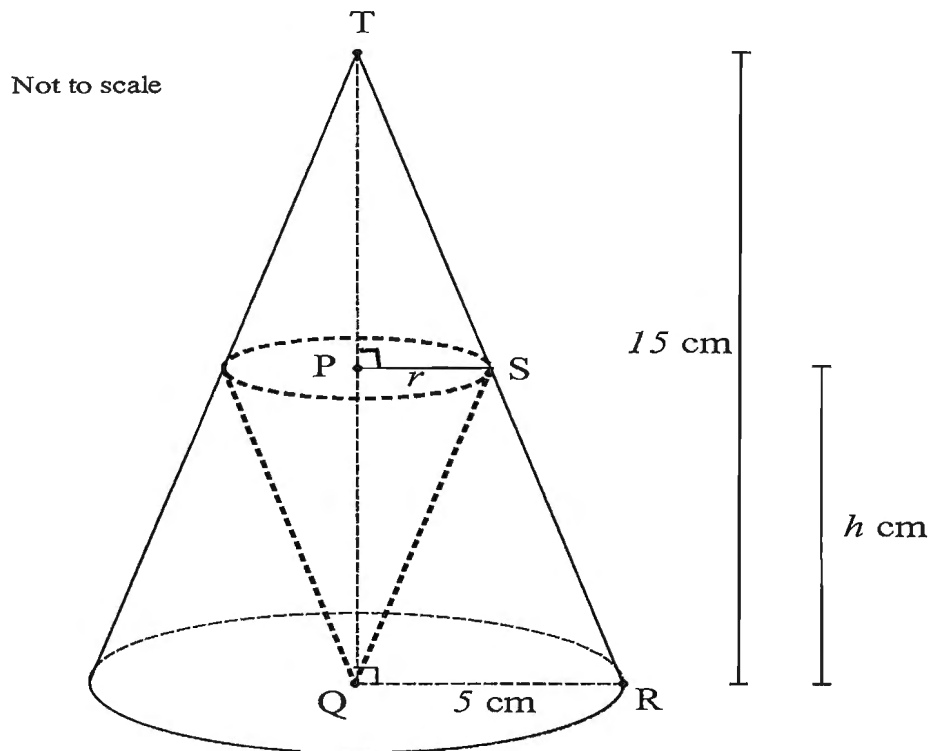
- | | | |
|-------|--|---|
| (i) | Calculate the exact length of BC . | 1 |
| (ii) | Show that the equation of BC is $2x - y + 5 = 0$. | 2 |
| (iii) | Find the exact area of triangle ABC . | 3 |

- b) Solve $x^2 - 2x - 1 < 0$. 3

Question 14 continues on the next page

Question 14 (continued)

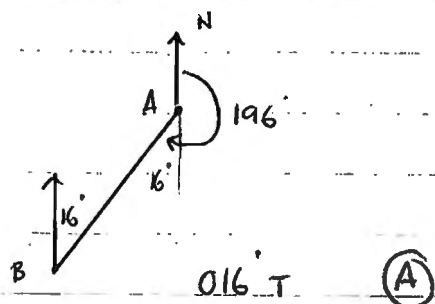
- c) The diagram shows two right cones. The small cone of radius $PS=r$ cm is enclosed within a larger cone of radius $QR=5$ cm, as shown. The larger cone has height 15 cm and the small cone has height h cm.



- | | | |
|-------|--|---|
| (i) | Prove that $\Delta TPS \parallel \Delta TQR$ | 2 |
| (ii) | Show that $h = 15 - 3r$ | 1 |
| (iii) | Find the radius of the small cone such that its volume is a maximum. | 3 |

End of Examination

mult. Choice



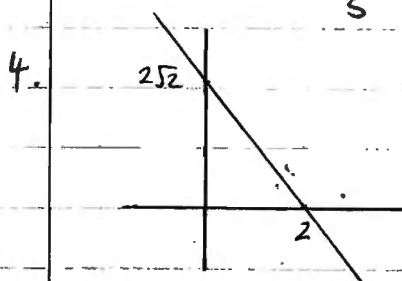
$$2. \frac{1}{5-\sqrt{3}} \times \frac{5+\sqrt{3}}{5+\sqrt{3}} = \frac{5+\sqrt{3}}{22}$$

3. $\alpha + \beta = \frac{-15}{5} = -3$

$$\alpha\beta = -\frac{3}{5}$$

$$\alpha^2 + 3\alpha\beta + \beta^2 = (\alpha + \beta)^2 + \alpha\beta$$
$$= (-3)^2 + \left(\frac{-3}{5}\right)$$

$$= \frac{42}{5}$$



$$m = \frac{-2\sqrt{2}}{2} = -\sqrt{2}$$

$$\tan \theta = -\sqrt{2}$$

$\therefore \angle \text{ of inclination} = 125^\circ$ (C)

5. Decreasing : $f'(x) < 0$

Conc. down : $f''(x) < 0$ (B)

6. $\frac{1}{\sin \theta} \cdot \frac{1}{\cos \theta}$

$\frac{\sin \theta}{\cos \theta}$

$$= \frac{1}{\sin \theta} \cdot \frac{1}{\cos \theta} \cdot \frac{\cancel{\cos \theta}}{\cancel{\sin \theta}}$$

$$= \frac{1}{\sin^2 \theta} = \operatorname{cosec}^2 \theta$$

7. $\tan \alpha = \frac{-1}{\sqrt{3}}$
(Related $\angle = 30^\circ$)
 $\alpha = 150^\circ, 330^\circ$ (C)

8. $\angle ACB = 180^\circ - \beta$
 $w^2 = u^2 + v^2 - 2uv \cos(180^\circ - \beta)$
 but $\cos(180^\circ - \beta) = -\cos \beta$
 $\therefore w^2 = u^2 + v^2 + 2uv \cos \beta$ (B)

9. Draw $y = 2 - |x|$
4 points of intersection
 $\therefore$ 4 solutions. (P)

$$\begin{aligned} 10. \quad f(x+1) &= x^2 - x + 1 \\ f(x) &= (x-1)^2 - (x-1) + 1 \\ &= x^2 - 2x + 1 - x + 1 + 1 \\ &= x^2 - 3x + 3 \end{aligned}$$

COVER	1
last time	1
Q1	3
12	3
13	4
14	4

Q11

$$\begin{aligned}
 a) \quad \frac{x+3}{5} - \frac{x-2}{4} &= \frac{4(x+3) - 5(x-2)}{20} \quad \leftarrow (1) \\
 &= \frac{4x+12 - 5x+10}{20} \\
 &= \frac{-x+22}{20} \quad \text{or} \quad \frac{22-x}{20} \quad \leftarrow (1)
 \end{aligned}$$

$$\begin{aligned}
 b) \quad |3x-2| &= 7 \\
 3x-2 &= 7 \quad \text{or} \quad -3x+2=7 \\
 3x &= 9 \quad \text{or} \quad -3x=5 \\
 x &= 3 \quad \leftarrow (1) \quad \quad \quad x = -\frac{5}{3} \quad \leftarrow (1)
 \end{aligned}$$

$$c) \quad 6x^2 + 11x - 10 = (3x+2)(2x+5) \quad \leftarrow (2)$$

[Correct terms, Signs wrong $\leftarrow 1$]

$$\begin{aligned}
 d) \quad y &= \sqrt{x^2-16} \\
 (i) \quad D: \quad x^2-16 &\geq 0 \\
 \therefore x &\leq -4, \quad x \geq 4 \quad \leftarrow (1)
 \end{aligned}$$

$$R: \quad y \geq 0 \quad \leftarrow (1)$$

$$\begin{aligned}
 (ii) \quad f(x) &= \sqrt{x^2-16} \\
 f(-x) &= \sqrt{(-x)^2-16} \\
 &= \sqrt{x^2-16} \\
 &= f(x) \quad \therefore \text{Even} \quad \leftarrow (1)
 \end{aligned}$$

$$\begin{aligned}
 e) \quad x^2+y^2 &\leq 9 \quad \text{or} \quad y \leq \sqrt{9-x^2} \\
 y &\geq x \\
 y &\geq 0
 \end{aligned}
 \left. \vphantom{\begin{aligned} x^2+y^2 &\leq 9 \\ y &\geq x \\ y &\geq 0 \end{aligned}} \right\} \begin{array}{l} \leftarrow \text{all three (2)} \\ \text{[Any two} \leftarrow 1 \text{]} \end{array}$$

$$\begin{aligned}
 f) \quad 4(1-\sin^2 \theta) - 6 \sin \theta - 6 &= 0 \quad \leftarrow (1) \\
 4 - 4\sin^2 \theta - 6 \sin \theta - 6 &= 0
 \end{aligned}$$

$$-4 \sin^2 \theta - 6 \sin \theta - 2 = 0$$

$$(\div -2) \quad 2 \sin^2 \theta + 3 \sin \theta + 1 = 0$$

$$(2 \sin \theta + 1)(\sin \theta + 1) = 0 \quad \leftarrow (1)$$

$$\sin \theta = -\frac{1}{2} \quad \text{or} \quad \sin \theta = -1$$

$$\underline{\theta = 210^\circ, 330^\circ, 270^\circ} \quad \leftarrow (1)$$

Q12.

$$a) \quad \frac{3^{2n} - 1}{6^n - 2^n} = \frac{(3^n - 1)(3^n + 1)}{2^n(3^n - 1)} \quad \leftarrow (1) \text{ factorise numerator and/or denominator}$$

$$= \frac{3^n + 1}{2^n} \quad \leftarrow (1) \text{ correct answer}$$

$$b) \quad i) \quad y = 4(1-x)^{-1}$$

$$y' = 4 \cdot -(1-x)^{-2} \cdot -1$$

$$= 4(1-x)^{-2} \quad \text{or} \quad \frac{4}{(1-x)^2}$$

$$ii) \quad y = 2x \cdot \sqrt{3x+1}$$

$$\left| \begin{array}{ll} u = 2x & u' = 2 \\ v = (3x+1)^{1/2} & v' = \frac{1}{2}(3x+1)^{-1/2} \cdot 3 \end{array} \right.$$

$$\frac{dy}{dx} = \cancel{2x} \times \frac{3}{\cancel{2}\sqrt{3x+1}} + \sqrt{3x+1} \times 2 = \frac{3}{2\sqrt{3x+1}}$$

$$= \frac{3x}{\sqrt{3x+1}} + 2\sqrt{3x+1}$$

(1) — Uses product rule

(1) — correct ans.

$$\left(\text{or } 3x(3x+1)^{-1/2} + 2(3x+1)^{1/2} \right)$$

(iii)

$$y = \frac{4x-1}{2x+1}$$

$$u = 4x-1$$

$$u' = 4$$

$$v = 2x+1$$

$$v' = 2$$

$$y' = \frac{vu' - uv'}{v^2}$$

$$= \frac{(2x+1) \cdot 4 - (4x-1) \cdot 2}{(2x+1)^2}$$

← (1) uses quotient rule

$$= \frac{8x+4-8x+2}{(2x+1)^2}$$

$$= \frac{6}{(2x+1)^2}$$

← (1) correct answer

c) $\lim_{x \rightarrow -2} \frac{x^3+8}{2x+4} = \lim_{x \rightarrow -2} \frac{(x+2)(x^2-2x+4)}{2(x+2)}$ ← (1) factorise numerator

$$= \frac{(-2)^2 - 2(-2) + 4}{2}$$

$$= \frac{4+4+4}{2}$$

$$= 6$$

← (1) correct answer

d) $m = \frac{dy}{dx} = \frac{2x-4}{2x+4}$
 $= \frac{2(1)-4}{2(1)+4}$ at $(1, -3)$
 $= -2$

← (1) correct gradient

$$y - y_1 = m(x - x_1)$$

$$y + 3 = -2(x - 1)$$

$$y + 3 = -2x + 2$$

$$y = -2x - 1$$

$$\text{or } 2x + y + 1 = 0$$

← (1) correct eqn.

e) $|2x-3| = 4x+1$

$$2x-3 = 4x+1$$

$$-4 = 2x$$

$$x = -2$$

but LHS = $|-7| = 7$ } not equal
 and RHS = -7

∴ not a solution

$$-2x+3 = 4x+1$$

$$2 = 6x$$

$$x = \frac{1}{3}$$

$$\text{LHS} = |-2 \cdot \frac{1}{3} + 3| = 2\frac{1}{3}$$

$$\text{RHS} = 2\frac{1}{3}$$

> equal

∴ Soln.

Solution: $x = \frac{1}{3}$ only

(1) for $x = \frac{1}{3}$

(1) for show and reject $x = -2$

Q13.

a)



$$\cos \theta = -\frac{1}{\sqrt{50}}$$

(1) for neg.

(1) for $\frac{1}{\sqrt{50}}$.

b)

$$\frac{x+4}{5} = \frac{12}{x-7}$$

$$(x+4)(x-7) = 5 \times 12$$

$$x^2 - 3x - 28 = 60$$

$$x^2 - 3x - 88 = 0$$

} (1)

← (1)

$$(x+8)(x-11) = 0$$

$$x = -8, 11 \quad \text{but } x > 0$$

$$\underline{x = 11}$$

← (1)

c)

$$\text{RHS} = A(x^2 + 4x + 4) + Bx + C$$

$$= Ax^2 + 4Ax + 4A + Bx + C$$

$$= Ax^2 + (4A+B)x + (4A+C)$$

$$\text{LHS} = 6x^2 + 0x - 11$$

$$\underline{A = 6}$$

$$4(6) + B = 0$$

$$4(6) + C = -11$$

$$\underline{B = -24}$$

$$\underline{C = -35}$$

← (1)

← (1)

← (1)

d)

For real roots, $\Delta \geq 0$

$$\Delta = b^2 - 4ac$$

$$= (-2)^2 - 4 \cdot 1 \cdot (3-k)$$

← (1)

$$= 4 - 12 + 4k$$

$$= 4k - 8 \geq 0 \quad \text{when } \underline{k \geq 2} \quad \leftarrow (1)$$

e)

$$y = \frac{1}{2}x^4 - 2x^3 + 2$$

$$y' = 2x^3 - 6x^2$$

$$y'' = 6x^2 - 12x$$

(i) For stationary points, $y' = 2x^3 - 6x^2 = 0$
 $2x^2(x-3) = 0$

$$x=0$$

or

$$x=3$$

$$y=2$$

$$y=-$$

$$(0, 2) \leftarrow (1)$$

$$(3, -11.5) \leftarrow (1)$$

(ii) At (0, 2):

$$y'' = 6(0)^2 - 12(0) = 0 \quad \therefore \text{Possible pt. of inflexion}$$

x	-1	0	1
y''	+18	0	-6

$$6(-1)^2 - 12(-1)$$

$$6(1)^2 - 12(1)$$

Since concavity change

(0, 2) is a horizontal pt. of inflexion
 $\leftarrow (1)$

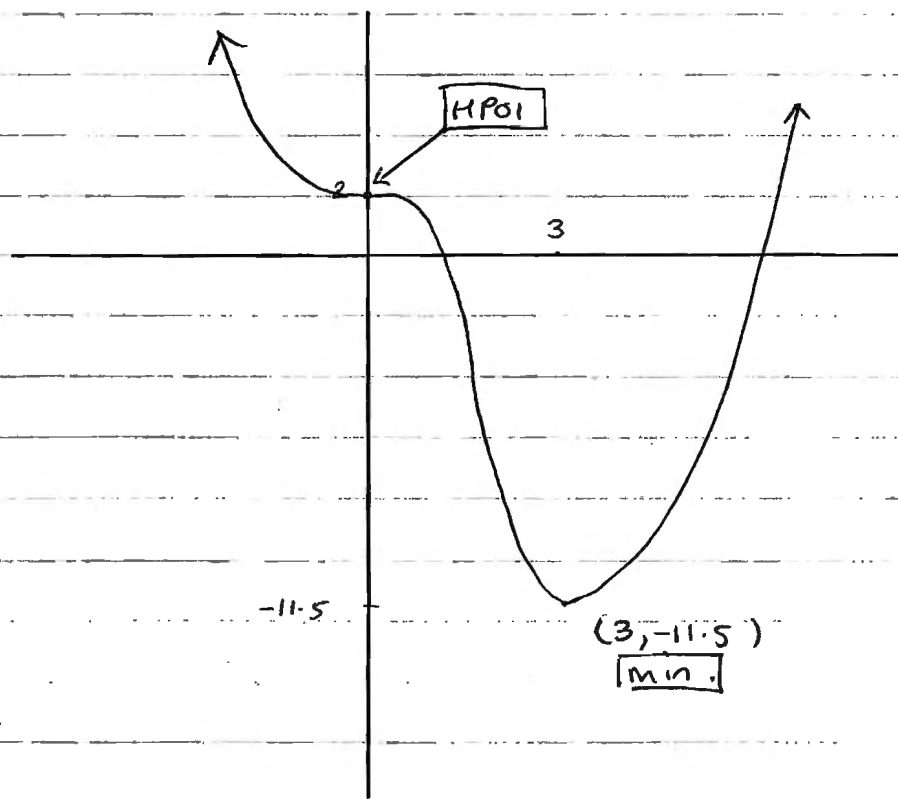
At (3, -11.5)

$$y'' = 6(3)^2 - 12(3) = +18$$

$$> 0$$

(3, -11.5) is a minimum turning point
 $\leftarrow (1)$

(iii)



(2)

b) $x^2 - 2x - 1 < 0$

Roots: $x^2 - 2x - 1 = 0$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{2 \pm \sqrt{(-2)^2 - 4(1)(-1)}}{2 \times 1}$$

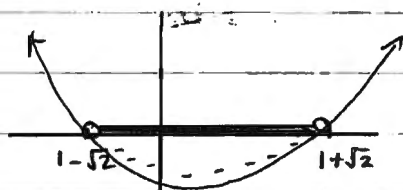
$$= \frac{2 \pm \sqrt{8}}{2} \quad \leftarrow 2\sqrt{2}$$

$$= \frac{2 \pm 2\sqrt{2}}{2}$$

$$= 1 \pm \sqrt{2}$$

$\leftarrow (1)$

Sketch



$\leftarrow (1)$

Soln: $\underline{1 - \sqrt{2} < x < 1 + \sqrt{2}}$

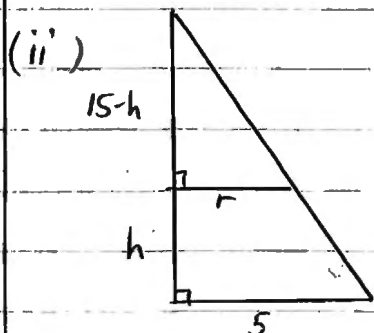
$\leftarrow (1)$

c) i) In $\triangle TPR$, $\triangle TQS$

$\angle TPR = \angle TQS$ (given, both 90°)

$\angle T$ is common

$\therefore \triangle TPR \parallel \triangle TQS$ (matching $\angle$ s equal) or (AA)



$$\frac{15-h}{15} = \frac{r}{5}$$

$$15r = 75 - 5h$$

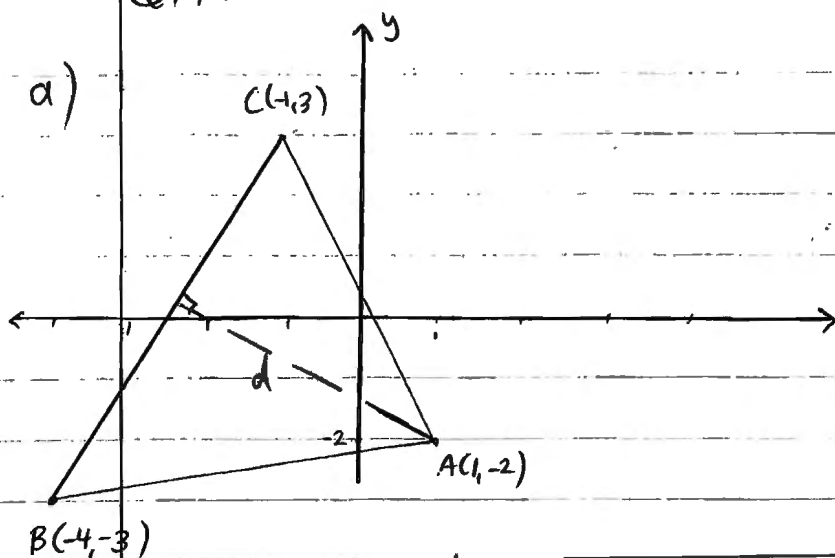
$(\div 5) \quad 3r = 15 - h$

$$\underline{h = 15 - 3r}$$

$\leftarrow (1)$

Q14.

a)



$$\begin{aligned}
 \text{(i)} \quad BC &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\
 &= \sqrt{(-1 + 4)^2 + (3 + 3)^2} \\
 &= \sqrt{9 + 36} \\
 &= \sqrt{45} \text{ or } 3\sqrt{5} \text{ units}
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad BC \text{ has } m &= \frac{y_2 - y_1}{x_2 - x_1} \\
 &= \frac{3 + 3}{-1 + 4} \\
 &= \frac{6}{3} \\
 &= 2
 \end{aligned}$$

and passes through (-1, 3) ← (1)

$$\begin{aligned}
 \therefore y - y_1 &= m(x - x_1) \\
 y - 3 &= 2(x + 1) \quad \leftarrow (1) \\
 y - 3 &= 2x + 2 \\
 2x - y + 5 &= 0
 \end{aligned}$$

$$\text{(iii)} \quad d = \frac{|2(1) - (-2) + 5|}{\sqrt{2^2 + (-1)^2}} = \frac{9}{\sqrt{5}} \quad \leftarrow (1) \quad \leftarrow (1)$$

$$\begin{aligned}
 \text{Area of } \triangle ABC &= \frac{1}{2}bh = \frac{1}{2} \times 3\sqrt{5} \times \frac{9}{\sqrt{5}} \\
 &= \frac{27}{2} \text{ units}^2 \quad \leftarrow (1)
 \end{aligned}$$

(iii)

$$V = \pi r^2 h \quad \text{but} \quad h = 15 - 3r$$

$$V = \pi r^2 (15 - 3r)$$

$$= 15\pi r^2 - 3\pi r^3 \quad \leftarrow (1)$$

For maximum volume, $V' = 0$ and $V'' < 0$

$$V' = 30\pi r - 9\pi r^2$$

$$= 3\pi r (10 - 3r)$$

$$= 0 \quad \text{when} \quad r \neq 0 \quad \text{or} \quad r = \frac{10}{3} \quad \leftarrow (1)$$

$$V'' = 30\pi - 9\pi(2r)$$

$$= 30\pi - 18\pi r$$

$$= 30\pi - 18\pi \left(\frac{10}{3}\right)$$

$$= 30\pi - 60\pi$$

$$< 0, \quad \text{as required}$$

$$r = \frac{10}{3} \text{ cm gives max. volume}$$

(1)